

SCAD Blended Cyber Operations Curriculum: 6-Month Linux-Centric Breakdown

This document provides the full, detailed breakdown of the 6-month (24-week) SCAD Apprenticeship Curriculum. The program is designed for **960 total hours** of intensive, hands-on training, with a mandatory focus on the Linux Operating System across all modules.

Program Structure and Time Allocation

The 24-week program is divided into three phases, ensuring a strong foundation before specialization and culminating in a comprehensive capstone project.

Phase	Duration (Weeks)	Total Hours (40 hrs/week)	Modules Covered	Focus
I: Foundation & Core Skills (SCAD-CORE)	4 Weeks	160 Hours	CORE101, CORE102, CORE103	Mandatory Linux, Networking, and Scripting Mastery
II: Specialization Track	18 Weeks	720 Hours	6 Specialization Units	Deep-dive into chosen domain (OCO, DFIR, SOC, or GRC)
III: Integrated Capstone (SCAD-CAP)	2 Weeks	80 Hours	CAP101, CAP102, CAP103	Real-world, end-to-end simulation and professional reporting
Total	24 Weeks	960 Hours		

Phase I: Foundation & Core Skills (SCAD-CORE) - 4 Weeks (160 Hours)

This phase is mandatory for all apprentices, ensuring a uniform, high-level proficiency in the operational environment.

Course Code	Course Title	Duration (Weeks)	Total Hours	Theory/Lecture (Hours)	Hands-on Lab (Hours)
CORE101	Linux Command Line & System Administration	1.5	60	15	45
CORE102	TCP/IP, Networking & Linux Network Tools	1.5	60	20	40
CORE103	Python & Bash for System Automation	1	40	10	30
Total	4 Weeks	160 Hours	45	115	

Phase II: Specialization Tracks - 18 Weeks (720 Hours)

Each specialization track is covered over 18 weeks, with each of the 6 units allocated 3 weeks for in-depth coverage and extensive lab work.

Programme: Offensive Cyber Operations (SCAD-OCO) - 18 Weeks

Week Range	Course Code	Course Title	Linux Integration Focus	Total Hours
Weeks 5-7	OCO101	Network & Web Reconnaissance Techniques on Linux	Using Linux tools (whois, dig, Nmap) for passive/active recon; scripting OSINT automation.	120
Weeks 8-10	OCO102	Scanning & Vulnerability Assessment with Kali Linux Tools	Hands-on use of Kali Linux, Nessus/OpenVAS deployment, and vulnerability scanning from a Linux host.	120
Weeks 11-13	OCO103	System Exploitation & Linux Privilege Escalation	Exploiting Linux misconfigurations (SUID, PATH), kernel exploits, and post-exploitation on Linux targets.	120
Weeks 14-16	OCO104	Web App Attacks & OWASP Top 10 using Linux Proxies	Using Burp Suite/OWASP ZAP on Linux, targeting Linux web servers, and command injection via Bash.	120
Weeks 17-19	OCO105	Maintaining Access & Covering Tracks on Linux Systems	Creating persistent backdoors (SSH keys, cron jobs), rootkits, and manipulating Linux logs.	120
Weeks 20-22	OCO106	Red Team Exercises (Linux-based)	Full-cycle red team engagement using a Linux C2 server, targeting a mixed Linux/Windows environment.	120

Programme: Digital Forensics & Incident Response (SCAD-DFIR) - 18 Weeks

Week Range	Course Code	Course Title	Linux Integration Focus	Total Hours
Weeks 5-7	DFIR101	Digital Evidence Concepts & Linux Forensics Principles	Linux file system structure (ext4, XFS), and proper acquisition of Linux memory/disk images using Linux tools (dd, dcfldd).	120
Weeks 8-10	DFIR102	Disk, File Systems & Data Recovery on Linux	Analyzing Linux file systems with Autopsy/TSK, recovering deleted files, and examining inode structures.	120
Weeks 11-13	DFIR103	Linux Malware Investigation & Triage	Static/dynamic analysis of Linux binaries, using tools like strace, ltrace, and sandboxing Linux malware.	120
Weeks 14-16	DFIR104	Packet Capture, Logs & Network Evidence on Linux	Analyzing network traffic with Wireshark/tshark on Linux, and advanced log analysis of system (syslog, journald) and application logs.	120
Weeks 17-19	DFIR105	Containment, Investigation & Reporting in a Linux Environment	Using Linux IR tools (GRR, osquery) for live response, isolating compromised Linux hosts, and creating forensic timelines.	120
Weeks 20-22	DFIR106	Forensic Lab Exercises (Linux Focus)	Practical exercises on real-world Linux compromise scenarios, including root access and data exfiltration.	120

Programme: SOC & Blue Team Operations (SCAD-SOC) - 18 Weeks

Week Range	Course Code	Course Title	Linux Integration Focus	Total Hours
Weeks 5-7	SOC101	Security Operations Center Overview & Linux Hardening	Introduction to SOC roles, and mandatory Linux system hardening (kernel tuning, SELinux/AppArmor, systemd security).	120
Weeks 8-10	SOC102	SIEM, Logs, & Event Correlation on Linux	Configuring log collection (rsyslog, filebeat) from Linux hosts, writing custom log parsers, and monitoring Linux audit logs.	120
Weeks 11-13	SOC103	Identifying Threats & Anomalous Behavior on Linux	Writing detection rules (YARA, Sigma) for Linux-specific threats, and using Linux utilities to identify suspicious processes.	120
Weeks 14-16	SOC104	Response, Escalation & Containment Procedures on Linux	Scripting automated containment actions (firewall rules, process killing) on Linux hosts, and using SSH for secure remote investigation.	120
Weeks 17-19	SOC105	Firewalls, IDS/IPS & Endpoint Protection on Linux	Deploying and managing Linux-based security tools (Snort/Suricata, OSSEC/Wazuh), and configuring host-based firewalls.	120
Weeks 20-22	SOC106	Simulated SOC Environment & Reporting (Linux-based)	Operating a full SOC stack (ELK/Splunk) deployed on Linux, and responding to simulated attacks against Linux servers.	120

Programme: GRC & Cyber Risk Operations (SCAD-GRC) - 18 Weeks

Week Range	Course Code	Course Title	Linux Integration Focus	Total Hours
Weeks 5-7	GRC101	Governance, Risk & Compliance Basics	Understanding how GRC frameworks (ISO 27001, NIST) apply to Linux-based infrastructure and open-source environments.	120
Weeks 8-10	GRC102	Risk Assessment & Mitigation Techniques for Linux	Identifying and assessing risks related to Linux configuration, patching, and third-party software dependencies.	120
Weeks 11-13	GRC103	Policy Creation & Control Implementation on Linux	Translating security policies into enforceable Linux configurations (e.g., password complexity, account lockout, file integrity monitoring).	120
Weeks 14-16	GRC104	Audit Support & Regulatory Requirements (Linux Focus)	Using Linux tools (OpenSCAP, Lynis) to automate compliance checks against standards like CIS Benchmarks for Linux.	120
Weeks 17-19	GRC105	Risk Reporting & Metrics from Linux Systems	Generate compliance reports by extracting and analyzing security data from Linux logs and configuration files.	120
Weeks 20-22	GRC106	Scenario-Based Risk & Compliance Exercises (Linux Audit)	Conducting a full security audit of a deployed Linux server against a specific compliance standard.	120

Phase III: Integrated Capstone (SCAD-CAP) - 2 Weeks (80 Hours)

This final phase integrates all learned skills in a high-fidelity, simulated operational environment.

Week Range	Course Code	Course Title	Description	Total Hours
Week 23	CAP101/102	Project Planning & Integrated Simulation	Apprentices define project scope, set up the target Linux environment, and execute the full-cycle simulation (GRC, OCO, SOC, DFIR).	40
Week 24	CAP103	Professional Reporting & Executive Briefing	Focus on creating the final professional report (in DOC format) that details findings, vulnerabilities, the incident timeline, and mitigation recommendations, followed by a presentation.	40
Total	2 Weeks			80 Hours

Detailed Weekly Breakdown

Phase I: Foundation & Core Skills (SCAD-CORE) - Weeks 1-4

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
1	CORE101	Linux Command Line & System Administration I	10	30	40
		- Shell basics (Bash), navigation, file system hierarchy (FHS)			
		- File manipulation (cp, mv, rm, find, grep, sed, awk)			
		- User and group management, permissions (chmod, chown)			
2	CORE101/102	Linux Admin II & Network Fundamentals I	15	25	40
		- Process management (ps, top, kill), service control (systemd)			
		- Package management (apt/dpkg), environment variables			
		- TCP/IP stack, OSI model, basic routing concepts			
3	CORE102	Networking & Linux Network Tools	15	25	40
		- Subnetting, DNS, DHCP, ARP			
		- Network analysis (tcpdump, Wireshark basics, netstat, ss)			
		- Firewall configuration (iptables/ufw), secure remote access (SSH, key management)			
4	CORE103	Python & Bash for System Automation	10	30	40
		- Bash scripting fundamentals (loops, conditionals, functions)			
		- Python basics (data types, control flow)			

		- Automating administrative tasks and log parsing with Python/Bash			
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Phase II: Specialization Tracks - Weeks 5-22

Offensive Cyber Operations (SCAD-OCO)

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
5	OC0101	Recon & Footprinting I (Passive)	10	30	40
		- OSINT techniques, domain/IP lookups (dig, whois)			
		- Scripting OSINT data collection with Python/Bash			
6	OC0101	Recon & Footprinting II (Active)	10	30	40
		- Network scanning with Nmap (advanced flags, scripting Nmap output)			
		- Service enumeration and banner grabbing on Linux targets			
7	OC0101	Recon & Footprinting III (Web)	10	30	40
		- Web enumeration (DirBuster/Gobuster on Linux), virtual hosts			
		- Footprinting cloud environments			
8	OC0102	Vulnerability Analysis I (Scanning)	10	30	40
		- Deploying and running Nessus/OpenVAS on a Linux host			
		- Interpreting scan results and false positives			
9	OC0102	Vulnerability Analysis II (Manual)	10	30	40
		- Manual vulnerability identification on Linux services (e.g., SSH, FTP)			
		- Configuration review of Linux systems (CIS Benchmarks)			
10	OC0102	Vulnerability Analysis III (Reporting)	10	30	40
		- Vulnerability management lifecycle and reporting			
		- Prioritization and risk scoring			
11	OC0103	Exploitation I (Metasploit & Payloads)	10	30	40
		- Using Metasploit Framework on Kali Linux			
		- Generating Linux-compatible payloads (msfvenom)			
12	OC0103	Exploitation II (Linux System Exploits)	10	30	40
		- Exploiting common Linux vulnerabilities (e.g., outdated services)			
		- Shell access and TTY spawning techniques			
13	OC0103	Exploitation III (Privilege Escalation)	10	30	40
		- Linux PrivEsc techniques (SUID, PATH, cron jobs, kernel exploits)			

		- Automated PrivEsc tools (LinPEAS)			
14	OC0104	Web & Application Hacking I (Injection)	10	30	40
		- SQL Injection and Command Injection targeting Linux web servers			
		- File Inclusion (LFI/RFI) attacks leading to RCE on Linux			
15	OC0104	Web & Application Hacking II (OWASP Top 10)	10	30	40
		- Cross-Site Scripting (XSS), CSRF, and broken authentication			
		- Using Burp Suite/OWASP ZAP on Linux for traffic analysis			
16	OC0104	Web & Application Hacking III (APIs)	10	30	40
		- API security testing and common API vulnerabilities			
		- Attacking containerized web applications on Linux (Docker)			
17	OC0105	Post-Exploitation I (Persistence)	10	30	40
		- Establishing persistence on Linux (SSH keys, cron jobs, systemd services)			
		- Lateral movement techniques within a Linux network			
18	OC0105	Post-Exploitation II (Data Exfiltration)	10	30	40
		- Data staging and exfiltration techniques (DNS, HTTP, ICMP tunneling)			
		- Interacting with Linux filesystems and databases			
19	OC0105	Post-Exploitation III (Covering Tracks)	10	30	40
		- Manipulating Linux logs (syslog, journald, history)			
		- Anti-forensics techniques on Linux systems			
20	OC0106	Red Team Exercises I (Planning & C2)	10	30	40
		- Red Team methodology, planning, and threat emulation			
		- Setting up a Linux Command and Control (C2) server (e.g., Covenant, Empire)			
21	OC0106	Red Team Exercises II (Execution)	10	30	40
		- Executing a full simulated attack against a target environment			
		- Bypassing common Linux security controls			
22	OC0106	Red Team Exercises III (Reporting & Debrief)	10	30	40
		- Professional reporting and executive summary creation			
		- Debriefing and handover to Blue Team			

Digital Forensics & Incident Response (SCAD-DFIR)

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
5	DFIR101	DF Fundamentals I (Principles & Linux Filesystems)	10	30	40
		- Digital evidence concepts, chain of custody			
		- Linux file system structure (ext4, XFS, Btrfs) and metadata			
6	DFIR101	DF Fundamentals II (Acquisition)	10	30	40
		- Proper acquisition of Linux disk images (dd, dcfldd)			
		- Memory acquisition on Linux (LiME, fmem)			
7	DFIR101	DF Fundamentals III (Triage & Volatility)	10	30	40
		- Linux live response and triage using standard Linux tools			
		- Introduction to Volatility for Linux memory analysis			
8	DFIR102	Storage Analysis I (Disk & File Systems)	10	30	40
		- Using The Sleuth Kit (TSK) and Autopsy on Linux			
		- Examining inode structures and deleted file recovery			
9	DFIR102	Storage Analysis II (Data Recovery)	10	30	40
		- File carving techniques (foremost, scalpel)			
		- Analyzing swap space and hibernation files on Linux			
10	DFIR102	Storage Analysis III (Timeline)	10	30	40
		- Creating and analyzing super timelines (log2timeline/plaso)			
		- Interpreting MACB times on Linux			
11	DFIR103	Malware Analysis I (Triage & Static)	10	30	40
		- Malware triage process, indicators of compromise (IOCs)			
		- Static analysis of Linux binaries (strings, file, objdump)			
12	DFIR103	Malware Analysis II (Dynamic)	10	30	40
		- Dynamic analysis on Linux (strace, ltrace, GDB)			
		- Setting up a safe Linux sandbox for malware execution			
13	DFIR103	Malware Analysis III (Reverse Engineering)	10	30	40
		- Introduction to Linux reverse engineering tools (IDA Pro/Ghidra)			
		- Analyzing common Linux malware families (e.g.,			

		cryptominers, backdoors)			
14	DFIR104	Network Forensics I (Packet Capture)	10	30	40
		- Advanced packet capture with tcpdump and tshark on Linux			
		- Protocol analysis and stream reconstruction			
15	DFIR104	Network Forensics II (Log Analysis)	10	30	40
		- Advanced analysis of Linux system logs (syslog, journald)			
		- Analyzing web server logs (Apache/Nginx) for attacks			
16	DFIR104	Network Forensics III (Wireless & Cloud)	10	30	40
		- Wireless network forensics basics on Linux			
		- Cloud log analysis (AWS/Azure) and correlation with Linux hosts			
17	DFIR105	Incident Response I (Preparation & Detection)	10	30	40
		- IR planning, runbooks, and team roles			
		- Using osquery/GRR for continuous monitoring and detection on Linux			
18	DFIR105	Incident Response II (Containment & Eradication)	10	30	40
		- Isolating compromised Linux hosts (firewall rules, network segmentation)			
		- Eradication and secure rebuilding of Linux systems			
19	DFIR105	Incident Response III (Recovery & Lessons Learned)	10	30	40
		- System recovery, validation, and post-incident hardening			
		- Documentation and lessons learned			
20	DFIR106	Forensic Lab Exercises I (Case Study)	10	30	40
		- Full-cycle investigation of a Linux web server compromise			
		- Evidence handling and chain of custody documentation			
21	DFIR106	Forensic Lab Exercises II (Malware Incident)	10	30	40
		- Investigating a Linux-based ransomware or cryptominer incident			
		- Creating a detailed forensic report			
22	DFIR106	Forensic Lab Exercises III (Insider Threat)	10	30	40
		- Investigating an insider data exfiltration scenario on a Linux workstation			
		- Mock court presentation of findings			

SOC & Blue Team Operations (SCAD-SOC)

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
5	SOC101	SOC Fundamentals I (Roles & Processes)	10	30	40
		- SOC structure, roles (Tier 1, 2, 3), and common workflows			
		- Threat intelligence integration			
6	SOC101	SOC Fundamentals II (Linux Hardening I)	10	30	40
		- Linux system hardening (CIS Benchmarks, kernel tuning)			
		- Secure configuration of SSH and user accounts			
7	SOC101	SOC Fundamentals III (Linux Hardening II)	10	30	40
		- Mandatory access control (SELinux/AppArmor) implementation			
		- File integrity monitoring (FIM) on Linux (AIDE, Wazuh)			
8	SOC102	Monitoring & Logging I (Linux Log Sources)	10	30	40
		- Understanding Linux log sources (syslog, journald, auditd)			
		- Configuring log collection (rsyslog, filebeat) from Linux hosts			
9	SOC102	Monitoring & Logging II (SIEM Deployment)	10	30	40
		- Deploying and configuring a Linux-based SIEM (e.g., ELK Stack, Splunk Free)			
		- Data normalization and parsing			
10	SOC102	Monitoring & Logging III (Event Correlation)	10	30	40
		- Writing correlation rules for Linux events			
		- Alert tuning and false positive reduction			
11	SOC103	Threat Detection I (Rule Writing)	10	30	40
		- Writing detection rules for Linux-specific threats (YARA, Sigma)			
		- Mapping detections to MITRE ATT&CK framework			
12	SOC103	Threat Detection II (Anomalies)	10	30	40
		- Identifying suspicious processes and network connections on Linux			
		- Behavioral analysis and baseline creation			
13	SOC103	Threat Detection III (Hunting)	10	30	40
		- Proactive threat hunting methodologies on Linux systems			
		- Using osquery for deep system visibility			
14	SOC104	Incident Handling I (Triage & Containment)	10	30	40
		- Incident triage and severity scoring			

		- Automated containment scripting on Linux (firewall rules, process killing)			
15	SOC104	Incident Handling II (Investigation)	10	30	40
		- Secure remote investigation via SSH and jump boxes			
		- Root cause analysis on Linux systems			
16	SOC104	Incident Handling III (Reporting & Automation)	10	30	40
		- Incident reporting and communication protocols			
		- SOAR (Security Orchestration, Automation, and Response) basics			
17	SOC105	Defensive Tools I (Firewalls & IDS)	10	30	40
		- Advanced Linux firewall configuration (nftables)			
		- Deploying and tuning Snort/Suricata on Linux			
18	SOC105	Defensive Tools II (Endpoint Protection)	10	30	40
		- Deploying and managing Linux-based EDR/HIDS (e.g., Wazuh, OSSEC)			
		- Antivirus and malware scanning on Linux			
19	SOC105	Defensive Tools III (Vulnerability Management)	10	30	40
		- Patch management strategies for Linux environments			
		- Integrating vulnerability scanners with SOC workflow			
20	SOC106	SOC Lab Exercises I (Attack Simulation)	10	30	40
		- Simulating a common Linux attack (e.g., SSH brute force) and detecting it in the SIEM			
		- Full incident response drill			
21	SOC106	SOC Lab Exercises II (Threat Hunting)	10	30	40
		- Proactive hunting for a specific threat actor technique on the lab network			
		- Creating a new detection rule based on the hunt			
22	SOC106	SOC Lab Exercises III (Reporting)	10	30	40
		- Generating daily/weekly SOC reports and metrics			
		- Presenting findings to a simulated management team			

GRC & Cyber Risk Operations (SCAD-GRC)

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
5	GRC101	GRC Fundamentals I (Concepts)	10	30	40
		- Governance, Risk, and Compliance definitions and interplay			
		- Key GRC frameworks (NIST CSF, ISO 27001, COBIT)			

6	GRC101	GRC Fundamentals II (Linux Context)	10	30	40
		- Applying GRC principles to open-source and Linux environments			
		- Understanding the legal and regulatory landscape (e.g., GDPR, HIPAA)			
7	GRC101	GRC Fundamentals III (Risk Culture)	10	30	40
		- Building a security-aware culture			
		- Stakeholder management and communication			
8	GRC102	Risk Management I (Assessment)	10	30	40
		- Risk assessment methodologies (qualitative vs. quantitative)			
		- Identifying Linux-specific threats and vulnerabilities (e.g., zero-day in kernel)			
9	GRC102	Risk Management II (Mitigation)	10	30	40
		- Risk treatment options (mitigate, transfer, accept, avoid)			
		- Developing mitigation strategies for Linux infrastructure			
10	GRC102	Risk Management III (Analysis)	10	30	40
		- Business Impact Analysis (BIA) and risk register maintenance			
		- Calculating risk exposure for Linux systems			
11	GRC103	Security Policies I (Creation)	10	30	40
		- Developing clear, enforceable security policies			
		- Policy mapping to technical controls on Linux (e.g., password policy)			
12	GRC103	Security Policies II (Implementation)	10	30	40
		- Translating policies into automated Linux configurations (e.g., Ansible/Puppet for hardening)			
		- Control implementation for access control and least privilege on Linux			
13	GRC103	Security Policies III (Review)	10	30	40
		- Policy review and maintenance lifecycle			
		- Exception handling and documentation			
14	GRC104	Compliance Operations I (Auditing)	10	30	40
		- Audit process and preparation (internal and external)			
		- Using Linux tools (OpenSCAP, Lynis) to automate compliance checks			
15	GRC104	Compliance Operations II (Regulatory)	10	30	40
		- Deep dive into a specific regulation (e.g., PCI DSS, NERC CIP) and its Linux requirements			
		- Evidence collection and documentation for audits			

16	GRC104	Compliance Operations III (Continuous Monitoring)	10	30	40
		- Implementing continuous compliance monitoring on Linux hosts			
		- Remediation tracking and corrective action plans			
17	GRC105	Reporting & Documentation I (Metrics)	10	30	40
		- Defining key risk indicators (KRIs) and key performance indicators (KPIs)			
		- Extracting security data from Linux logs and configuration files for metrics			
18	GRC105	Reporting & Documentation II (Risk Reporting)	10	30	40
		- Creating clear, concise risk reports for executive management			
		- Data visualization for GRC metrics			
19	GRC105	Reporting & Documentation III (Documentation)	10	30	40
		- Maintaining system security plans (SSPs) and control documentation			
		- Disaster Recovery/Business Continuity Planning			
20	GRC106	Practical GRC Lab I (Compliance Audit)	10	30	40
		- Conducting a full compliance audit of a Linux server against CIS Benchmarks			
		- Generating a formal audit report with findings			
21	GRC106	Practical GRC Lab II (Risk Scenario)	10	30	40
		- Scenario-based risk assessment (e.g., new open-source software deployment)			
		- Developing a full risk treatment plan			
22	GRC106	Practical GRC Lab III (Policy Implementation)	10	30	40
		- Writing a new security policy and implementing the corresponding technical controls on a Linux test environment			
		- Presenting the policy and implementation plan			

Phase III: Integrated Capstone (SCAD-CAP) - Weeks 23-24

Week	Course Code	Topic Focus	Theory (Hrs)	Lab (Hrs)	Total (Hrs)
23	CAP101/102	Integrated Simulation Execution	10	30	40
		- Environment setup and final policy review (GRC)			
		- Full-scale attack execution (OCO) and real-time monitoring (SOC)			
		- Live response and evidence collection (DFIR)			
24	CAP103	Final Reporting & Executive Briefing	10	30	40
		- Forensic and penetration test report generation			

		- Executive briefing and final presentation of findings and recommendations			
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